Pierre Onghena

Deep Learning & 3D Computer Vision

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AI researcher that focuses on deep learning and 3D computer vision for point cloud perception. First-author publications, with prior industry experience in LiDAR technology, applied ML, and consulting. Now looking to translate knowledge into solutions.

EXPERIENCE

Mines Paris – PSL Oct 2023 – Dec 2026
PhD Candidate *Paris, France*

- Develop novel methods for 3D point cloud analysis at the Centre for Mathematical Morphology, integrating geometric priors to improve model performance and robustness (PyTorch).
- Design, run, and evaluate large-scale training pipelines on the Jean Zay HPC cluster using Slurm job scheduling.
- Mentor master's-level engineering students in project-based research internships.

ESCP Business School 2024 – 2026
Teaching Assistant *Paris, France*

- Led lectures for Bachelor in Management students, driving participation and conceptual understanding.
- Taught mathematics in the first semester to equip students with analytical and quantitative reasoning skills.
- Taught statistics in the second semester to prepare students to reason about uncertainty in complex problems.

IBM May – Sep 2023
Data Engineer *Brussels, Belgium*

- Adapted large language models to domain-specific document collections, improving automated handling of customer inquiries for corporate and public-sector clients.
- Contributed to consulting engagements and technical tenders within IBM's global technology practice, translating ML expertise into client-facing value propositions.

The Cross Product Jul 2022 – Mar 2023
R&D Engineer *Fontainebleau, France*

- Built automated semantic segmentation of 3D point clouds for rail, road, and industrial infrastructure at a deep-tech startup, collaborating closely with technical and business teams.
- Delivered results that led to a peer-reviewed publication and conference presentation.

EDUCATION

MSc Artificial Intelligence, Maastricht University 2020 – 2022
MSc Artificial Intelligence, Systems & Data, Paris Dauphine – PSL Jan – Jun 2022
BSc Business & Information Systems Engineering, KU Leuven 2019 – 2020
BSc Business Engineering, KU Leuven 2016 – 2019

PUBLICATIONS

P. Onghena, S. Velasco-Forero, B. Marcotegui **BMVC Oral**
"GEKey: Learning Geodesic Eccentric 3D Keypoints via Self-Supervision" *Lancaster, UK, 2026*

M. Bastico, **P. Onghena**, D. Ryckelynck, et al. **ISBI Oral**
"LmPT: Conditional Point Transformer for Anatomical Landmark Detection on 3D Point Clouds" *London, UK, 2026*

P. Onghena, S. Velasco-Forero, B. Marcotegui **3DV**
"MorphoSkel3D: Morphological Skeletonization of 3D Point Clouds for Informed Sampling" *Singapore, 2025*

P. Onghena, L. Gigli, S. Velasco-Forero **ICCVW**
"Rotation-invariant Hierarchical Segmentation on Poincaré Ball for 3D Point Cloud" *Paris, France, 2023*

J. Aechtner, L. Cabrera, D. Katwal, **P. Onghena**, et al. **WCCI**
"Comparing User Perception of Explanations Developed with XAI Methods" *Padua, Italy, 2022*

EXTRACURRICULAR

- **Languages:** English (Full Professional Proficiency) · French (Full Professional Proficiency) · Dutch (Native)
- **Rail Open Lab:** participation in building the bridge between research and its implementation in the railway sector.
- **Sailing Scout Leader:** coordinated sailing weekend activities and summer camps for secondary school students.
- **Catamaran Instructor:** completed didactic and technical modules to teach sailing to beginners.
- **Sports:** golf, rowing, tennis and field hockey player.